

Materials in the DARJa repository

Data and data preparation

The R file `darja_data3.RData`, which contains an object called `d`, has information on geographical IDs, geographical coordinates, feature type, feature, and feature values. A corresponding tab-delimited text file with the same rows is offered solely for those who do not wish to engage with R; it's called `darja_data3.txt`, and is given as a zip file.

Files used in the preparation of `darja_data3.RData` are: `DARJa_data_preparation2.R`, `data-small-final2.csv`, `Cintaksis.xlsx`, `Fonetika.xlsx`, `Leksika.xlsx`, `Morfologiya.xlsx`. The last four were prepared by a team at Kazan Federal University and downloaded by us on 2022-03-30 (they are still available at <https://kpfu.ru/atributivnaya-baza-dannyh-russkih-govorov-269324.html> as of 2025-07-21). The Kazan files have IDs for locations, coordinates in an unidentified system, and features and their values. Using the script `DARJa_data_preparation2.R` we extracted this information and combined it with information on geographical coordinates gathered by ourselves through georeferencing the generic DARJa map—information contained in `data-small-final2.csv`. None of the files `DARJa_data_preparation2.R`, `data-small-final2.csv`, `Cintaksis.xlsx`, `Fonetika.xlsx`, `Leksika.xlsx`, `Morfologiya.xlsx` are used in subsequent analyses, but they are supplied in order to document the nature of the original data and details of the data extraction. Note that locations with the IDs 628 and 2329 do not have proper numbers on the DARJa maps, and their map numbers are given as 0 or -2 in the Kazan files, even if the files do have features and feature values associated with them in those files. They are considered spurious and were deleted from our data, which then has 4193 rather than 4195 locations.

Computing linguistic distances

The R file `linguistic_distance_matrix4.RData`, which contains an R object called `m`, contains linguistic distances among the 4193 locations. Row and column names correspond to the IDs, so they are sequences from 1 to 4195 with 628 and 2329 missing. The distances were computed using `linguistic_distances_darja2.R`. That script also contains a toy example for demonstration of how the calculation works. Because of constraints on the size of files at GitHub, access to the linguistic distance matrix is provided as a link in the file `distance_matrices_links.txt`.

Computing geographical distances

The R file `geographical_distance_matrix3.RData`, which contains an R object called `mgeo`, contains geographical distances among the 4193 locations. Row and column names correspond to the IDs, so they are sequences from 1 to 4195 with 628 and 2329 missing. The distances were computed using `geographical_distance_matrix.R`. Because of constraints on the size of files at GitHub, access is provided as a link in the file `distance_matrices_links.txt`.

Clusterization and cluster validation

We produced ultrametric trees based on the distance matrix, using all methods available in the `hclust()` function of base R, except 'single', 'WPGMC', and 'UPGMC', which put nearly all locations in one big cluster even when $k = 5$, for instance. The method used was: 'ward.D', 'ward.D2', 'complete', 'average' (same as UPGMA), and 'mcquitty' (same as WPGMA). The trees are in the files `WardD.RData`, `WardD2.RData`, `complete.RData`, `UPGMA.RData`, `WPGMA.RData`, and the objects contained in the files have the same names as the files minus the suffix (so: `WardD`, `WardD2`, etc.). Clusterization was carried out by `clusterization_darja.R`.

Experiments with cluster validation were directed at finding some quantitative criteria for deciding between the 5 methods and some optimal number of clusters. All experiments are carried out using `optimal_number_clusters4.R`. The first section of the script implements using the Adjusted Rand index (ARI) to quantify the fit between the distribution of feature values over locations (dialects) and a given clusterization. Output of this is in `optimal_number_clusters2.txt`. The second section does silhouette scores on the clusterizations. The third section does stability scores, output of which is in `stability_scores.txt`. A final section contains some qualitative notes made during inspection of clusters in their geographical setting using the `dmap()` script (for which see below).

Maps

The `dmap()` function of the `dmap.R` script serves to produce different kinds of maps, some coming with an accompanying text file: custom maps of locations clustered according to different methods and different numbers of clusters, maps indicating geographical cluster outliers, a generic map showing only locations, a map of biomes, and custom maps of the distribution of individual clusters or cluster values.

It requires the files `darja_data3.RData`, `linguistic_distance_matrix4.RData`, `WardD.RData`, `WardD2.RData`, `WardD2`, `complete.RData`, `UPGMA.RData`, `WPGMA.RData`, `cities.RData`, `biomes_locs.txt`, and `locations.R`. A separate file with instructions is provided as `dmap_manual.pdf`.

Biomes

The script `biomes.R` reads biome data from Olson & Dinerstein (1998) posted at <https://www.worldwildlife.org/publications/terrestrial-ecoregions-of-the-world#:~:text=There%20are%20867%20terrestrial%20ecoregions,conserve%20biodiversity%20around%20the%20world>. The files are in https://files.worldwildlife.org/wwfcmprod/files/Publication/file/6kcchn7e3u_official_teow.zip. Unzipped, this will yield a folder, which should be renamed to `official_teow` and be placed as a subfolder of the folder where `biomes.R` is found. The script `biomes.R` identifies the biome associated with each location in DARJa and creates a data frame with IDs, coordinates, and a number corresponding to the biome of each location. This is output to the file `biomes_locs.txt`. The numbers are associated with descriptions in the little file called `key_biomes98.txt`. Subsequently, `biomes.R` creates a

tab-delimited file called `biomes3.txt` with a row per pair of locations giving the location IDs, the geographical and linguistic distances, and a 0 for ‘same’ and 1 for ‘different’ biome. This large file is shared via a link in the `biome_data_link.txt`; its contents is further analyzed using the script `analyze_biome_data.R`. This script investigates the effect of biomes on linguistic distance when controlling for geographical distance. Results are in `biome_results.txt` and the `biome_difference.jpeg` plot.

Rivers

Using `rivers.R`, the files `rivers_m.txt` and `rivers_l.txt` were produced. The script and the text files, as zipped files, are provided. The files list all pairs of locations that are less than 300 km apart with their linguistic and geographical distances, indicating the number of rivers or lakes that separate them. The river data come from <https://www.naturalearthdata.com/downloads/> via the interface of the `rnaturalearth` R package. The file name elements `rivers_m` and `rivers_l` refer respectively to the medium and large-scale data of `naturalearth`. ‘Medium’ means 1:50 m(illion) (medium resolution, relatively few rivers) and ‘large’ 1:10 mill (higher resolution, relatively many rivers). The data in `rivers_m.txt` and `rivers_l.txt` are read and analyzed using `analyze_river_data.R`. We report only on the higher-resolution data since the lower-resolution data are impoverished and also do not add to the findings.

Administrative divisions

All relevant code is in `administrative_units.R`. This creates a file called `admin_units.txt`, which has a row per pair of locations giving the location IDs, the geographical and linguistic distances, and a 0 for ‘same’ and 1 for ‘different’ oblast as well as a 0 for ‘same’ and 1 for ‘different’ rajon. This large file is shared via a link in `admin_divisions_data_link.txt`.

All predictors combined

The code for the combined model of all predictors and the barplot of their relative contributions is in `combined_model.R`.